AWS-Part-1

AWS EC2 Instance

Virtual server

ECC - Elastic Cloud compute Instance

Mean Scalable

Cloud based

Calculate cost & Pay per use only

→ EC2 is provides an virtual server
means it has it's own - CPU
- RAM
- Storage [EBS]
- OS [AMI]
- Cloud Network
- IP
- Security group i.e firewall
- Key pair for secure login.

→ After Launching the Instance, next step would be connect It

Mean accesing your virtual server to control it,
- run commands
- host
- deploy

TO connect → ssh command is used

→ ssh [secure shell] is an ~~cor~~ command used to securely connect to remote server over a network using encryption

Syntax: ssh [option] user@host

↑ root, ubuntu, ec2-user

↑ IP address or domain name

identify file from this directory

Eg: ssh -i ~/Download/Jenkins-server-key.pem ubuntu@16.170.224.195

EC2 instance creation

1.] Requirement Gathering
2.] Key pair
3.] Security Group
4.] Launch the Instance
5.] Connect using ssh

I.] Requirement Gathering

- Which OS?
- How much size needed
- How much CPU compute needed
- Speed
- Used for which stage [Development, QA, Test, Staging, Production]

should focus on the naming convention!

II.) Key pair

- should not used same key pair for every instance
- should also not use unique key pair for every new instance
- Group it & assign a single key pair to a single stage. ~~for each~~ every environment.

Naming convention: projectname/ - regionname (Env) - [optional stage name]

key are region specific

III.] Security Group [Firewall]
[Which user are given permission to enter & leave]
basically rules.

Security Group

Inbound rules | Outbound rules

- Give tag to security group too to make more sense. ~~RA +~~

→ Using EC2 user data script, we can run a program at Launch of Instance once only called bootstrap code.
→ used to install software, update, download common files.

IV.] Launching the Instance:

Name & Tags

Key	Value	Resource Type
Name	web01	Instance, Volumes & Network Interface
Project	Mos O	
Environment	Dev	For all
Owner	DevOps Team	

V.] Connect using ssh

Notes

When instance is powered-OFF
- Public IP is lost
- Private IP remains the same

IP [Internet Protocol address]

- IPV4
- IPV6
- where it is going.
- contains the address of where the data is going

TCP

- Delivery system
- How data is delivered
- connection establishment
- in order
- without error

→ If you want a static IP address GTO to Elastic IPs [5 GTiven Free]
↳ so public IP is never changed.
- Public IP assigned to your instance only

How to Scale up/down?

→ Stop the instance
→ GTO to instance type in Launch Instance
→ Change the Instance type.

AWS CLI:- [Command Line Interface]

→ Accessing the AWS account on your terminal is possible using AWS CLI

→ You can do each & every service of AWS on your terminal with AWS CLI commands.

→ You can find those commands on AWS documentation or Letting VIE UTP T

→ Now you need to give your username & password to your terminal to connect the AWS account.
↳ Using Access Key!
↳ Also aws should be install at your OS.

EC2 [Elastic Cloud Compute]

→ EC2 Instance type:-

m5.2xlarge

- m → Instance class
- 5 → generation
- 2xlarge → Size of EC2

Type
- Compute Optimized (C)
- General Purpose (t,m)
- Memory [RAM] [volatile] [r,x,z,u]
- Storage [Harddisk] [Involatile] [i,d,h,im]
- HPC [p,g,inf itrn]
- Accelerated Computing

→ You can connect to EC2 instance, direct from AWS console using EC2 Instance connect.

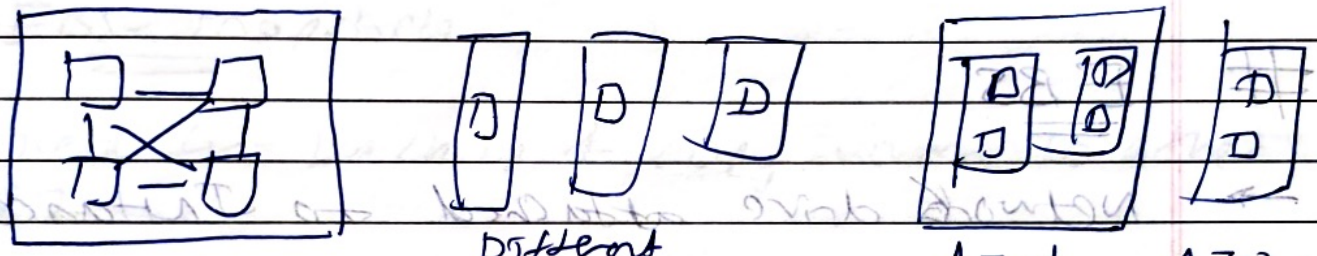
→ EC2 Purchasing Options
- On demand
- Reserved
  - Reserve Instance for 1/3 years
  - Charged whether used or not
- Savings
  - You define how much you are planning to spend on AWS for 1/3 years
- Spot Instances
  - Cheapest
  - You can lose instance at any time
  - Not suitable for critical jobs
- Dedicated Host & Instance
  - Reserves Entire physical server
  - Used for License purpose
- Capacity Reservation

Spot Fleet → EC2 on demand instance combined with spot instances

- Used for multiple AZs
- Non-critical task – Batch Jobs
  - CI/CD pipelines

EC2 Placement Group

Cluster | Spread | Partition



Reserve On-Demand Instance

- No time commitment

Elastic Network Interface [ENI]

→ A card which consist of all network info → IP address
→ Public, Private & Optional one

→ Suppose if one EC2 instance crashes, using Lambda + CloudWatch, automatically we can attach ENI to 2nd EC2
→ So no downtime, no lag.

EC2 States

Stop	Terminate	Hibernate
- Data on EBS is kept intact for next start	- EBS deleted	- in-memory [RAM] IP preserved - Starts faster like it was never shutted down - No Looting ~~time~~ time - Condn → Root EBS must be encrypted.

EBS